

Thinking Bidirectionally: A Reasoning and Self-Correction Approach for Text-Based Event Prediction with Large Language Models

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Abstract

Using Web Mining and Content Analysis to find and understand clues from the massive amount of unstructured text online is very important for predicting future events and providing early risk warnings in important fields like finance and public safety. While Large Language Models (LLMs) exhibit potential in processing and understanding text, current text-based event prediction faces two primary challenges: first, an insufficient utilization of potential information within the text, such as causal relationships and latent associations, and second, limited predictive reliability constrained by issues like the LLM's own ability and hallucinations. To address these challenges, we propose a novel event prediction framework, Bidirectional Reasoning with Self-Correction (BRSC). BRSC comprises two complementary reasoning dimensions: temporal deductive reasoning, which analyzes the trajectory of historical events along the timeline to enable accurate trend extrapolation, and synchronic associative reasoning, which deeply mines details and latent connections from documents within a specific time window to extended semantic information. In addition, we use a self-correction mechanism that identifies and rectifies potential hallucinations and errors during the reasoning process. Extensive experiments on international relations event prediction demonstrate that BRSC achieves significant improvements over several leading LLM-based methods.

CCS Concepts

• Computing methodologies → Information extraction.

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1 Introduction

The World Wide Web serves as an unprecedented real-time sensor of societal dynamics, yielding a continuous deluge of heterogeneous data from online news, social media, and institutional reports. Distilling this vast data stream into predictive models of future events, a task known as event prediction [3, 48], represents a grand challenge at the intersection of temporal modeling and causal inference. Success in this endeavor provides invaluable anticipatory intelligence, enabling proactive decision-making in high-stakes domains such as public safety [24, 43], finance [34], and public healthcare [25, 38, 40].

A main approach in event prediction uses structured data, such as knowledge graphs [9, 10, 24]. The basic idea of these methods is a trade-off: giving up the rich details of raw text to get a clear structure and logical consistency. Even though these methods are well-developed, they face built-in limitations from this data structuring process. Most real-world event information is found in unstructured text formats, like online news articles and official reports. As a result, changing the text into structured formats unavoidably loses important meaning and contextual links, leading to significant information loss [34]. This shows the strong need for methods that can work directly on the original source texts.

To circumvent the information loss from structured conversion, some research has shifted towards direct text-based prediction. However, many current approaches, which primarily leverage Large Language Models (LLMs), engage in only a superficial analysis. They use methods like summarization or multi-agent systems for a

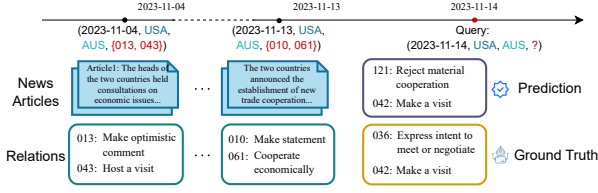


Figure 1: Event prediction example. Based on historical event tuples and news articles (left), the model predicts future event relations for a query. These predictions are then compared with the ground truth.

preliminary pass over the source documents but fail to deeply mine fine-grained information [12, 17], such as the associations between event elements and causal logic. Consequently, their performance improvements remain limited [13]. Furthermore, the promise of using LLMs for prediction is hampered by their inherent limitations. The model's propensity for hallucination, coupled with its failure to adhere to specified instructions and output formats, leads to unstable performance in complex reasoning, making it difficult to ensure the accuracy and consistency of the final predictions [40].

To overcome these aforementioned limitations, we introduce **Bidirectional Reasoning with Self-Correction (BRSC)**, a novel framework inspired by the human investigative process of "sequentially tracing evidence, associating related facts, and iteratively correcting conclusions". This method primarily consists of bidirectional reasoning and self-correction. The bidirectional reasoning part is divided into two complementary dimensions: **Temporal Deductive Reasoning (TDR)** and **Synchronic Associative Reasoning (SAR)**. TDR performs a longitudinal analysis, tracing event evolution along a timeline to extrapolate future trends. SAR conducts a detailed documents analysis, connecting disparate information within specific time windows to generate new and insightful predictions via semantic propagation. The **Self-Correction (SC)** module is designed to identify and modify model-induced hallucinations and errors that arise during both the reasoning and final prediction phases.

To summarize our main contributions:

- We introduce **BRSC**, a novel prediction framework designed for text-based event prediction that emulates the human "trace-associate-correct" reasoning model.
- We propose **TDR**, a method designed to analyze the trajectory of events in historical texts for the purpose of trend extrapolation.
- We propose **SAR**, a method designed to mine fine-grained details from texts within the same time window, enabling prediction through semantic propagation.
- Our experimental results demonstrate that the proposed method significantly outperforms existing approaches using LLMs on the task of text-based event prediction.

2 Related Work

2.1 Event Prediction

To understand and anticipate future event trajectories, research has long focused on the structured representation of event sequences

[31] and the task of event prediction. Early work utilized script narrative chains to structure events [7] and employed foundational language models to comprehend and predict event elements [11, 29]. Contemporary research in event prediction can be broadly categorized into three main streams. The first is event schema induction, which involves constructing and completing event frameworks [9, 18, 21, 22, 30]. The second focuses on knowledge graph completion, aiming to fill in missing elements within event-centric knowledge graphs [19, 24, 32, 39, 44]. The third encompasses question-and-answer (QA) prediction tasks, where predictions are formulated by constructing and answering questions about future event elements [12, 13, 17, 47, 50]. From a technical standpoint, knowledge graph prediction tasks have primarily relied on various neural network models, whereas text-based prediction research has burgeoned with the advent of LLMs.

Current text-based event prediction focuses on multi-agent methods and prediction system construction. Some studies [12, 13, 17] established "retrieval-prediction" processes, while MIRAI [43] utilized multi-agent document understanding. However, existing methods fail to fully mine profound textual knowledge or provide reasoning frameworks for accurate prediction, lagging behind human performance. Thus, novel methods for deeper text mining are essential to improve prediction accuracy.

2.2 Large Language Model Reasoning Enhancement

In order to make up for the logical fragility of large language models (LLMs) in complex reasoning tasks due to their probabilistic nature, reasoning enhancement technology has emerged. Its core is to optimize the intermediate process of model generation. The mainstream technology paradigm is evolving in three directions: 1. Striving to optimize the internal reasoning structure of the model and improve the logic and exploratory nature of its thinking process. [4, 5, 8, 15, 35, 36, 41, 46] 2. Using tool learning methods to call on external senses to break through the performance bottleneck of the model itself. [26, 28, 42] 3. Using the strategy of structural decomposition and multi-agent collaboration, complex problems are broken down into subtasks and handled collaboratively by professional agents [14, 37, 45, 49].

Event prediction features rich spatiotemporal relationships and logical details, offering significant potential for deep reasoning and mining. However, these elements remain underexplored. Developing novel methods to explicitly leverage these connections could thus lead to substantial performance gains.

3 Task Definition

We define an event as consisting of a quadruple $e = (t, s, r, o)$, where t is the daily timestamp, $s, o \in \mathcal{E}$ are the subject and object, and $r \in \mathcal{R}$ is their relationship (event content). Let $E = \{e_1, e_2, \dots, e_n\}$ be the set of historical events, E_t denotes events at time t , and all historical events occurring before t can be recorded as $E_{<t}$. A news article is $d = (t, c)$, where c represents the text content of news. The original news article library of all recorded events is recorded as $\mathcal{D}$, and the set of news articles before time t is recorded as $\mathcal{D}_{<t}$.

The primary objective of this research is to predict the set of entire relations R_q that will hold between a subject entity s and an

object entity o at a specified future time t' . We formalize this task as follows: given the historical context provided by both events $E_{<t}$ and articles $D_{<t}$ and a query of the form $q = (t', s, ?, o)$, the ultimate goal is to infer the set of relations R_q that will be instantiated at time t' .

4 Method

By emulating the human reasoning path of 'trace, associate, and correct,' our proposed method (BRSC) establishes a multi-stage process to perform predictions. The process commences with news article retrieval, which filters news texts based on the subject and object entities from the given query $q = (t', s, ?, o)$ to compile a relevant collection of articles $\{D_1, D_2, \dots, D_{t'-1}\}$, providing a sufficient evidence base for the prediction. Subsequently, this collection of articles is processed by three core modules: Temporal Deductive Reasoning (TDR), Synchronic Associative Reasoning (SAR), and Self-Correction (SC). Figure 2 illustrates how these four modules collaborate, and we will detail each module in the following sections.

4.1 Temporal Deductive Reasoning

This module is responsible for the trace analysis within our process, analyzing the trajectory of events along the timeline to extrapolate future trends.

4.1.1 Key Point Summary and Event Tuple Construction. Given that the volume of historical news articles for a query can be substantial, a key challenge is to distill the essential event development trends. We first filter the news article set D_t at time point t , removing redundant articles and irrelevant information. The filtered articles are then summarized to create a text summary set A_t . From these summaries, we extract key information and formalize it into a four-tuple event description, yielding the event set E_t . This process provides the historical event set $E_{<t}$ and the summary set $A_{<t}$ required for our analysis.

4.1.2 Temporal Deductive Analysis. Given the historical event set $E_{<t}$ and summary set $A_{<t}$, we employ LLMs to identify underlying evolutionary trends, event distribution patterns, and causal associations.

- **Evolution trend analysis:** The LLM is tasked with constructing the evolutionary timeline of events from historical data, summarizing the overall trajectory, and analyzing key event points by incorporating background knowledge to identify their likely driving factors. Based on the established trends, the model then proposes several potential future developments, ranked by their probability.
- **Distribution pattern analysis:** This analysis examines the historical frequency and distribution of each event relation type. We investigate event occurrence rates within specific time periods to identify both periodic patterns and anomalous phenomena. By summarizing these temporal characteristics, we can infer the most probable event types for the given prediction time.
- **Causal association analysis:** This analysis discovers causal relationships by inspecting key correlating factors in the historical event set $E_{<t}$ and summary set $A_{<t}$. We then leverage

these identified causal laws, in conjunction with pre-query events, to derive potential future events.

We synthesize these findings by prompting the LLM with the analytical information $E_{<t}$ and $A_{<t}$. The model's output provides the primary evidence for the final prediction.

4.1.3 Trend Extrapolation Prediction. The analysis of evolutionary trends, distribution patterns, and causal associations from the LLM are used as a prompt. By integrating this information with the relation types in CAMEO, we infer the prediction result R^{TDR} via temporal deductive reasoning.

4.2 Synchronic Associative Reasoning

This module aims to generate divergent possibilities from a single time period by analyzing combinations of textual clues to infer novel insights, thereby expanding the scope of event prediction.

4.2.1 Articles Screening and Summarization. Historical articles retrieved for a query often cover extensive time spans (months or even years), yet our data possesses daily granularity. Consequently, analyzing every single time point is impractical and would introduce significant noise into the prediction process. To address this, we group articles into temporal intervals for analysis. For a given interval $T = \{t_{start}, \dots, t_{end}\}$, the corresponding article set is D_T . We employ LLMs to evaluate and rank the articles in D_T based on four criteria: predictive significance, informational richness, relevance and coverage of the period's key events. The top-ranked set of documents is denoted as D'_T . Finally, each article in D'_T is concisely summarized, preserving its essential information, to produce the summary set A'_T .

4.2.2 Association Reasoning. Associative reasoning simulates the human analytical model of combining multiple clues, as seen in investigation. By deeply mining textual details, it establishes profound semantic connections between seemingly unrelated entities and events, forming a more comprehensive and insightful understanding of complex situations.

We perform associative reasoning on the summary set A'_T derived from a single time interval. This process involves analyzing combinations of abstracts from A'_T to synthesize novel, predictively significant insights that emerge from integrating their details, thereby generating several potential inferences. These generated inferences are then validated by an LLMs to filter out any that are illogical or redundant with existing information. Finally, validated inferences constitute the inference set P_T .

This process is iterative. The newly generated inferences in P_T are incorporated into a feedback loop for subsequent rounds of reasoning. In each new round, we jointly sample from both the original summary set A'_T and the updated inference set P_T . These samples are combined to perform further associative reasoning, and any resulting novel inferences are validated and used to augment the set P_T . The associative reasoning process is illustrated in Figure 3.

4.2.3 Semantic Propagation Prediction. Associative reasoning expands the key semantics of the original text, enriching it with additional details. These resulting inferences are used to construct a prompt, which yields the prediction set R^{SAR} .

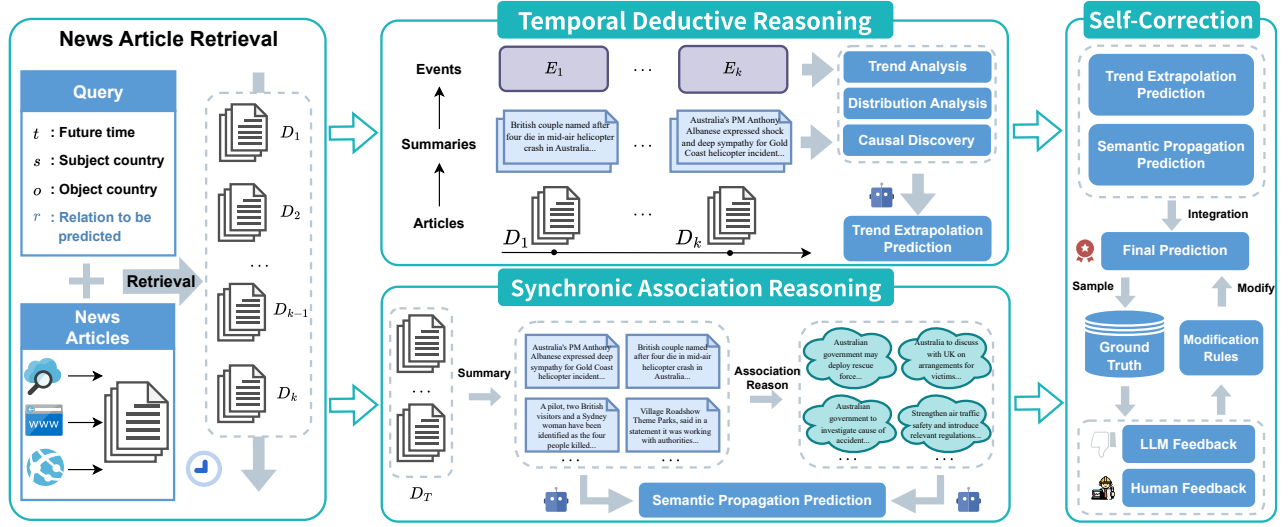


Figure 2: BRSC architecture. The left section retrieves news documents based on a query. The upper TDR module extracts event tuples for prediction, while the lower SAR module performs associative reasoning on summaries. On the right, the self-correction module aggregates predictions and refines them using ground truth feedback.

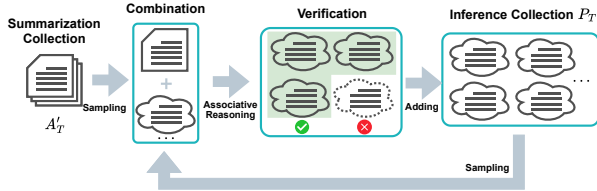


Figure 3: Associative reasoning process. An initial inference set, derived and validated from summary combinations, is iteratively expanded by sampling from both summaries and existing inferences for further relational deduction.

4.3 Self-Correction

This module then verifies and corrects the initial predictions, yielding a more accurate final result.

4.3.1 Error Type Analysis. Combining TDR and SAR yields the overall prediction $R_q = R_{TDR} \cup R_{SAR}$. Through sample analysis involving LLMs and human feedback, we identified three common error types: (1) **Format Errors**, where codes are undefined or violate CAMEO standards; (2) **Semantic Errors**, where codes contradict event meanings; and (3) **Hallucinatory Repetition**, involving duplicated codes, prompt examples, or excessive output.

4.3.2 Error Modification. Our correction methodology comprises two approaches. First, for general formatting errors, we employ targeted cleaning functions based on identified common error patterns. Second, to address semantic errors and hallucinatory repetitions, we compared a small sample of the model’s predictions against the ground truth. Through a collaborative analysis by human experts and the LLMs, we identified the root causes of these errors and

formulated correction strategies. These strategies are encoded into prompts, enabling the LLMs to self-correct and generate optimized predictions.

Semantic errors primarily stemmed from the LLMs’ misalignment between relation codes and their meanings. We mitigated this by supplementing prompts with CAMEO code documentation. Conversely, hallucinatory repetitions often appeared as the recycling of examples or meaningless output, particularly in smaller open-source models. This was addressed by instructing the LLM to strictly prohibit example duplication and filter for meaningful results.

Human feedback is strictly confined to the initial prompt engineering and sample analysis phases. Regarding sample analysis, we empirically identified a 5% sampling rate as the optimal threshold based on tests from 1% to 10%. Our results show diminishing returns beyond this point: the F1-score improved significantly from 72.4% (at 1%) to 85.1% (at 5%), with a negligible gain to 85.3% at 10%. Consequently, the subsequent self-correction operates as a fully automated pipeline by the LLM, requiring no further human intervention.

5 Experiments

5.1 Experimental Setup

Dataset. We validate our approach using the MIRAI [43] and GDELT-500 [20, 23] datasets. MIRAI is a recently constructed event prediction dataset derived from the Global Database of Events, Language, and Tone (GDELT). GDELT-500 is a benchmark dataset in the fields of knowledge graph completion and event prediction, which we have partially modified for our task by retrieving related historical documents from the web and selecting a suitable set of queries for validation.

Table 1: Statistics of the MIRAI and GDELT-500 datasets.

Statistic	MIRAI	GDELT-500
Date Range	2023.1.1–2023.11.30	2015.4.1–2016.3.31
News Articles	296,630	189,743
Historical Events	152,500	119,388
Queries	705	420

Table 1 presents the statistics for the two datasets. Historical events and news articles form the basis for prediction, while the query set—rigorously filtered and multi-source validated for accuracy—serves to evaluate the performance of event prediction methods.

5.1.1 Compared Approaches. We conducted our experiments on a set of 705 queries using *Qwen2.5-7B-Instruct* as the foundation model. The LLM’s temperature was set to 0.4 for all experiments, and we report the average results over 3 independent runs to ensure stability. The specific method of comparison is as follows:

- **Direct IO:** Directly prompt the LLM with the query. This will test the model’s capacity to predict events using only its pre-trained knowledge.
- **CoT [36]:** This approach involves providing basic task steps to instruct the LLM in completing the event prediction task.
- **ToT [41]:** Instead of being confined to the single, linear path of CoT, this approach utilizes a tree structure to simultaneously explore multiple different reasoning branches.
- **ReAct [42, 43]:** This is a multi-agent tool-learning approach that completes tasks via a cyclical “Think-Act-Observe” process.
- **BRSC:** Our approach, which uses bidirectional reasoning for document analysis and a self-correction mechanism to ensure reliable results.

5.1.2 Evaluation Metrics. The CAMEO framework uses two-digit codes for first-level relations. Second-level relations branch from these, adding a third digit to create a more specific, three-digit code. The final output is a set of relations R_q , consisting of three-digit codes that correspond to second-level CAMEO event types [43]. See Appendix C for event code details.

We evaluated the performance of our relational predictions using macro-averaged Precision, Recall, and F1 scores. To achieve this, the metrics were first computed for each query individually against the ground truth and then averaged across all queries. For the binary and quadratic classes predictions, we additionally employed Kullback-Leibler (KL) Divergence to measure the difference between the predicted and true event distributions. For specific calculation methods, please refer to the Appendix D.

5.2 Main Results

We evaluated our approach against baselines across six LLMs, comprising four open-source models (*Qwen2.5-7B-Instruct* [33], *Llama-3.1-8B-Instruct* [27], *internlm3-8b-instruct* [6], *Mistral-7B-Instruct-v0.2* [16]) and two commercial models (*GPT-4o* [1], *qwen-plus* [2]). Open-source models were deployed via vLLM (v0.9.1) on a dual NVIDIA RTX 4090 (24GB) system running Ubuntu 22.04. Tables 2 and 3 present the results, leading to the following conclusions:

5.2.1 Overall Performance Comparison. Our approach significantly outperforms baselines across most settings. Performance on MIRAI generally exceeds GDELT-500, attributed to the latter’s older events and lower text quality, highlighting the impact of data source reliability. Specifically, **our method achieves the lowest KL Divergence** in Binary and Quad class prediction tasks, closely matching the ground-truth distribution. For first-level and second-level prediction tasks, despite ReAct’s occasional higher Precision, **our method consistently secures superior overall F1 scores.**

5.2.2 Model Performance Comparison. While commercial models generally perform better, **our method significantly boosts open-source models.** For instance, using *Qwen2.5-7B* on MIRAI, our approach improved the Binary F1 score by 13% over ReAct, reaching 81.00%. This demonstrates our method as a general framework for structured prediction, enabling smaller open-source models to effectively handle complex tasks without relying solely on large-scale architectures.

5.2.3 Approach Performance Comparison. Approaches relying solely on internal knowledge (Direct, CoT) prove insufficient for complex event prediction. While the tool-augmented ReAct framework offers high Precision, its low Recall limits the overall F1 score. In contrast, **our BRSC approach resolves this trade-off by substantially boosting Recall while maintaining high Precision, resulting in a superior F1 score.** For second-level prediction, our method achieves over 40% Recall on numerous models—surpassing others by more than 10 percentage points. This enhanced recall ensures broader coverage of potential scenarios, making it highly effective for decision support.

5.2.4 Performance Analysis of Complex Tasks. Our approach’s advantage grows with task complexity. While all methods degrade on harder tasks (Binary to Second-level), baselines (Direct, CoT) suffer sharp losses on first- and second-level tasks. In contrast, **BRSC’s performance degrades far more gradual**, amplifying its relative lead where challenges are greatest. This underscores our framework’s superior robustness and scalability for complex, structured prediction.

5.2.5 Special Findings. We observed a notable anomaly where *internlm3-8b-instruct* suffered a precipitous performance decline on first- and second-level tasks using ReAct, primarily due to its inability to effectively utilize retrieval tools. This indicates that Agent-based capabilities are not uniformly robust across LLMs, a limitation our approach successfully circumvents.

5.3 Ablation Study

We conducted an ablation study to quantify the contribution of each component within our BRSC framework: the TDR, SAR, and SC modules. By systematically removing each module from the full model, we measured its respective impact on performance. Table 4 summarizes these results, and all of them are tested for statistical significance (p -value ≤ 0.01) using the t-test.

5.3.1 w/o TDR. Removing the TDR module causes the most significant performance degradation. As the primary module capturing temporal dynamics and event trends, TDR is critical for overall performance. Its removal eliminates the ability to reason over the

Table 2: Performance comparison of 5 prediction approaches across 6 LLMs on the MIRAI dataset. For metrics with an up-arrow (↑), higher is better, and for those with a down-arrow (↓), lower is better. The best score for each model is indicated in bold, * means statistically significant results (p -value ≤ 0.01) using the paired-t-test compared with the best baseline.

Type	Model	Approach	Binary		Quad		First-level			Second-level		
			F1 ↑	KL ↓	F1 ↑	KL ↓	Pre. ↑	Rec. ↑	F1 ↑	Pre. ↑	Rec. ↑	F1 ↑
Open-Source	Qwen2.5-7B-Instruct	Direct	58.66	10.04	39.42	13.42	26.98	18.68	14.86	13.81	6.58	6.59
		CoT	66.00	8.61	46.30	12.27	32.08	15.20	16.39	10.71	5.72	5.82
		ToT	67.66	8.17	45.56	12.18	31.21	15.63	17.41	11.85	7.51	7.44
		ReAct	67.56	6.91	51.03	9.92	36.09	36.54	29.26	25.02	28.96	20.85
		BRSC	81.00*	1.12*	64.56*	2.74*	35.62	68.07	38.11*	19.09	52.02	22.64*
	Llama-3.1-8B-Instruct	Direct	59.66	9.80	40.40	13.63	23.18	11.97	11.93	9.95	5.57	4.75
		CoT	62.66	9.42	41.76	13.51	28.83	13.78	14.24	11.36	6.55	6.39
		ToT	63.66	9.20	43.83	13.14	29.33	13.85	14.36	11.69	6.57	6.47
		ReAct	70.00	7.39	46.43	12.23	45.62	22.69	23.88	18.68	18.31	12.98
		BRSC	77.16*	2.63*	59.33*	6.15*	33.96	41.27	29.68*	15.35	24.65	13.66*
	internlm3-8b-instruct	Direct	66.66	8.63	45.20	12.83	31.50	6.34	9.38	5.91	3.08	3.73
		CoT	66.33	8.60	44.93	12.62	30.00	12.64	13.87	12.56	6.09	6.69
		ToT	65.33	8.83	44.26	12.89	30.00	12.39	13.67	11.36	5.46	5.90
		ReAct	66.67	8.63	32.23	16.89	6.00	3.63	3.86	1.60	1.93	1.36
		BRSC	73.22*	3.83*	57.22*	5.98*	36.18	49.38	29.53*	13.95	42.16	15.26*
	Mistral-7B-Instruct-v0.2	Direct	65.00	8.66	43.60	12.65	32.71	13.04	15.14	10.10	3.71	4.83
		CoT	70.66	7.11	46.68	11.95	30.66	10.51	11.99	7.21	4.63	4.08
		ToT	69.33	7.30	47.85	11.77	27.70	12.63	13.55	5.78	3.76	3.44
		ReAct	63.77	8.49	42.96	12.61	36.97	21.74	20.32	23.87	12.34	11.67
		BRSC	70.44	5.45*	52.16*	8.35*	29.61	30.19	22.57*	12.38	14.17	9.04
Commercial	Qwen-Plus	Direct	73.66	4.03	54.83	8.68	29.80	17.47	18.07	11.92	9.81	8.14
		CoT	78.33	1.42	53.38	8.88	32.91	19.80	20.37	17.34	11.82	10.81
		ToT	78.99	1.28	54.49	8.47	32.41	20.73	21.05	16.56	12.04	10.95
		ReAct	83.66	1.12	64.09	6.20	50.47	48.91	41.47	35.81	41.83	31.49
		BRSC	83.66*	1.07*	66.85*	2.36*	46.41	61.28	46.63*	33.71	51.67	34.69*
	GPT-4o	Direct	83.33	2.57	64.70	5.99	29.13	31.74	25.36	10.94	13.43	9.75
		CoT	81.00	2.58	62.02	5.69	31.40	38.13	28.62	12.02	17.78	10.72
		ToT	82.33	2.34	63.41	5.20	31.68	38.81	29.21	12.49	17.95	10.95
		ReAct	82.66	2.12	64.62	5.31	46.05	52.14	40.85	34.52	47.46	32.06
		BRSC	83.33*	1.11*	70.64*	2.12*	58.46	52.98	45.07*	48.15	53.85	40.25*

temporal dimension, leading to substantial information and performance loss.

5.3.2 w/o SAR. Removing the SAR module causes a moderate overall performance decline. For first-level and second-level tasks, precision slightly increases, but Recall drops sharply, lowering the F1 score. This is expected, as SAR supplements predictions by mining textual associations. Losing these supplementary insights naturally leads to a significant Recall drop, thereby impacting overall performance.

5.3.3 w/o SC. Ablating the SC module causes a comprehensive performance decline, most notably on simpler tasks. By correcting systematic errors, SC effectively boosts error-prone open-source models. The drop is steepest in Binary and Quad tasks, where removing SC’s noise-filtering capability causes a substantial decline.

5.4 Detailed Analysis

We further conduct a detailed analysis to identify the critical factors that influence each module’s behavior.

5.4.1 Analysis-1: Impact of Historical Event Quantity. To understand the impact of data volume on our trend-dependent TDR

module, we conducted an experiment on 42 queries with extensive event histories (>8,000 events). Using *Qwen2.5-7B-Instruct* and *GPT-4o*, we made predictions by randomly sampling different quantities of historical events. Each sampling condition was averaged over 20 runs. The results for first-level and second-level prediction are shown in Table 5.

Results show that increasing historical events significantly improves performance on first-level and second-level tasks, demonstrating the model’s ability to extract key events even near maximum context length. Our method is robust to data loss: omitting up to 50% of historical events degrades performance by less than 10%. Furthermore, larger historical data volumes enhance stability, evidenced by lower prediction variance across runs.

5.4.2 Analysis-2: Impact of Number of Inferences. We analyzed the impact of varying the number of inferences on model performance using 60 queries rich in historical articles. Figure 4 visualizes the results for the different models.

Results indicate that the number of inference impacts models differently. For open-source models, more inferences do not necessarily improve results, as excessive low-quality inferences can

Table 3: Performance comparison of 5 prediction approaches across 6 LLMs on the GDELT-500 dataset. For metrics with an up-arrow (↑), higher is better, and for those with a down-arrow (↓), lower is better. The best score for each model is indicated in bold, * means statistically significant results (p -value ≤ 0.01) using the paired-t-test compared with the best baseline.

Type	Model	Approach	Binary		Quad		First-level			Second-level		
			F1 ↑	KL ↓	F1 ↑	KL ↓	Pre. ↑	Rec. ↑	F1 ↑	Pre. ↑	Rec. ↑	F1 ↑
Open-Source	Qwen2.5-7B-Instruct	Direct	68.66	7.05	49.06	11.37	18.40	8.39	9.24	7.03	4.94	4.44
		CoT	70.66	5.86	51.26	11.04	16.96	14.91	11.04	11.13	7.65	6.18
		ToT	66.66	6.82	40.13	9.36	14.06	11.83	9.88	5.47	8.47	4.81
		ReAct	72.66	4.40	55.86	8.42	15.23	15.44	12.72	9.50	12.65	8.76
		BRSC	76.44*	1.70*	56.53*	3.61*	27.02	70.67	33.18*	14.41	55.25	19.02*
	Llama-3.1-8B-Instruct	Direct	58.0	9.68	42.26	13.32	10.00	9.85	7.72	4.40	8.69	4.12
		CoT	63.33	8.36	48.26	11.70	14.90	10.97	9.56	6.43	5.91	5.00
		ToT	70.66	6.73	52.4	10.88	17.90	13.41	11.62	7.63	8.18	6.06
		ReAct	73.77	5.45	51.33	11.07	40.66	33.01	32.10	12.91	25.26	13.27
		BRSC	76.00*	2.95*	56.09*	5.42*	23.57	50.49	28.45	9.45	31.68	13.30
	internlm3-8b-instruct	Direct	69.33	7.17	50.78	11.35	18.00	7.79	8.63	3.50	6.97	3.61
		CoT	72.66	6.15	51.79	10.43	23.01	13.79	14.55	7.16	6.15	4.97
		ToT	70.44	6.28	53.73	10.86	21.00	11.81	12.57	6.43	5.48	4.35
		ReAct	72.66	6.25	31.4	16.47	4.00	2.40	2.66	1.33	2.25	1.36
		BRSC	73.33*	3.84*	54.73*	6.01*	21.84	52.31	24.96*	8.92	47.97	12.45*
	Mistral-7B-Instruct-v0.2	Direct	56.66	9.94	44.40	12.64	18.16	12.63	12.08	5.77	8.93	5.48
		CoT	67.77	5.46	47.92	9.45	21.21	23.13	14.99	5.15	9.52	5.18
		ToT	69.99	6.32	51.86	9.86	20.66	11.61	12.46	5.70	5.72	5.09
		ReAct	73.99	5.69	52.39	10.33	19.0	9.84	10.34	13.0	4.42	4.77
		BRSC	75.33*	5.36*	57.51*	8.81*	18.41	27.27	16.98*	8.42	14.29	7.25*
Commercial	Qwen-Plus	Direct	63.33	4.35	51.26	7.76	18.83	21.58	16.92	8.45	14.33	8.13
		CoT	69.33	4.31	54.44	7.54	22.5	23.50	19.42	12.55	18.86	12.06
		ToT	77.99	1.73	58.73	7.14	26.83	21.53	20.47	13.19	16.69	12.11
		ReAct	79.55	1.59	64.11	4.19	43.11	66.24	45.47	28.15	56.15	31.68
		BRSC	82.00*	1.43*	58.34	2.92*	36.76	70.81	43.58	26.29	59.87	31.90*
	GPT-4o	Direct	78.00	3.11	57.89	5.34	19.86	30.82	20.41	6.20	20.76	7.34
		CoT	78.66	2.46	59.79	4.53	22.47	40.04	24.83	8.59	15.43	7.95
		ToT	81.98	1.89	61.71	4.03	23.21	38.21	25.52	7.53	20.59	8.56
		ReAct	80.66	1.41	59.97	3.55	30.83	59.84	35.21	21.79	57.15	26.88
		BRSC	84.00*	0.84*	62.25*	2.66*	38.33	73.97	44.90*	23.68	55.96	28.31*

Table 4: Ablation results for BRSC (full approach) vs. variants without TDR, SAR, or SC. Bolded values in parentheses denote performance drops.

Approach	Binary	Quad	First-level	Second-level
	KL ↓	KL ↓	F1 ↑	F1 ↑
BRSC	1.12	2.74	38.11	22.64
w/o TDR	3.58	6.78	25.53	9.49
	(+2.46)	(+4.04)	(-12.58)	(-13.15)
w/o SAR	2.42	4.89	33.31	20.69
	(+1.30)	(+2.15)	(-4.80)	(-1.95)
w/o SC	4.14	6.54	33.46	19.01
	(+3.02)	(+3.80)	(-4.65)	(-3.63)

be misleading. In contrast, commercial models generally show performance proportional to inference count. Overall, our approach demonstrates considerable stability, maintaining consistent performance despite these variations.

5.4.3 Analysis-3: LLM Error Distribution and Self-Correction Analysis. We conducted three rounds of experiments on 705 data entries

Table 5: Impact of historical event frequency on prediction performance across different models. Values are presented as mean F1-score $\pm$ standard deviation (%).

Events	Qwen2.5-7B-Instruct		GPT-4o	
	First-level	Second-level	First-level	Second-level
100	34.46 $\pm$ 27.13	13.19 $\pm$ 12.05	40.04 $\pm$ 13.46	32.04 $\pm$ 9.72
1000	36.77 $\pm$ 12.18	15.94 $\pm$ 7.09	44.37 $\pm$ 7.33	35.92 $\pm$ 5.86
2000	37.33 $\pm$ 8.63	16.09 $\pm$ 4.03	47.24 $\pm$ 5.67	38.52 $\pm$ 3.62
4000	44.75 $\pm$ 7.79	19.20 $\pm$ 3.78	48.53 $\pm$ 4.85	40.37 $\pm$ 3.05
6000	47.18 $\pm$ 6.84	20.25 $\pm$ 3.18	49.61 $\pm$ 4.41	41.09 $\pm$ 2.86
> 8000	47.83 $\pm$ 5.86	21.58 $\pm$ 3.03	50.36 $\pm$ 4.30	41.71 $\pm$ 2.64

to analyze the performance of our self-correction module, with results presented in the Figure 5.

Results show significant error rate variation across models. Commercial models substantially outperform open-source ones, indicating a clear correlation between model scale and accuracy. Semantic errors were the most prevalent. While our self-correction method resolves over 90% of format and repetition errors, semantic errors

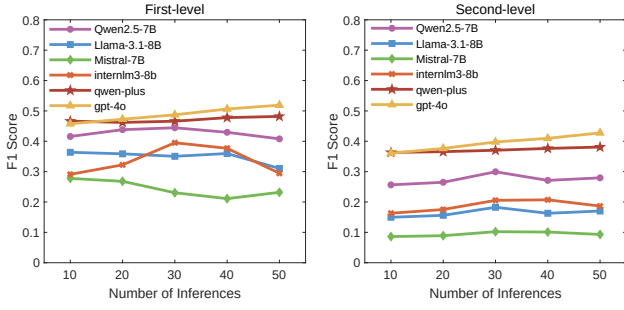


Figure 4: Performance comparison of various LLMs on first-level (left) and second-level (right) prediction tasks with varying numbers of inferences.

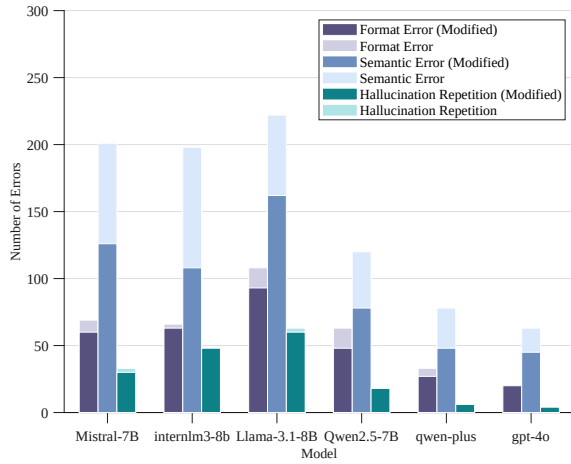


Figure 5: Analysis of error distribution and self-correction performance across various LLMs. The three colors categorize the errors: darker shades for corrected errors and lighter shades for uncorrected ones.

remain the hardest to fix. Nevertheless, we rectified over half of them, with commercial models performing exceptionally well.

5.4.4 Analysis-4: Resource Consumption and Computational Complexity. Due to the complex inference process of LLMs, we use resource consumption to measure the algorithm’s computational complexity. We measured inference latency and resource consumption on the MIRAI dataset across 6 models. Figure 6 illustrates the results averaged over 5 runs per query.

Compared to ReAct, agent-based BRSC has similar latency but superior token efficiency and performance, indicating that our approach does not significantly increase overhead relative to other agent-based methods. While agentic approaches generally incur higher costs than non-agent baselines, this is justified by their substantial improvements in accuracy.

6 Conclusion

In this work, we focus on comprehensively mining and analyzing rich information from the web text content to enhance the overall

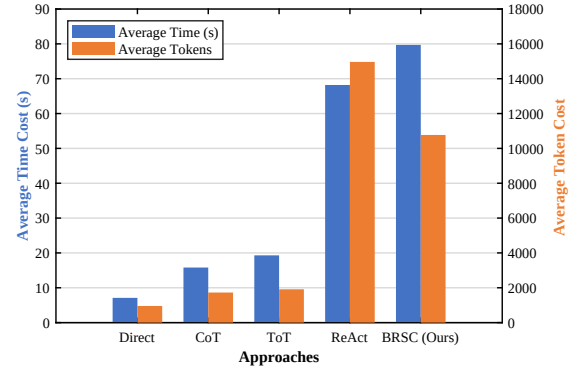


Figure 6: Comparison of average time and token costs across approaches. Blue bars indicate time (left axis), while orange bars denote tokens (right axis).

event prediction capabilities of LLMs. To this end, we now propose the BRSC framework, which systematically integrates two complementary dimensions of reasoning: the evolutionary trends of historical events and the rich details found within contemporaneous texts. By also leveraging the LLM’s own inherent self-correction capabilities to reduce errors, our framework is able to significantly enhance its predictive performance. Extensive experiments we conducted strongly demonstrate the superiority of our approach, while also revealing that substantial room for future research remains. Our future work will focus on further leveraging event relationships and adapting our inference logic to handle more complex prediction tasks over longer time spans.

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A Pseudocode

We provide pseudocode for each module to offer a clearer illustration of the algorithm.

Algorithm 1 Temporal Deductive Reasoning (TDR) Module

Require: Relevant news articles D_{relevant}
Ensure: Predicted relation set R_{TDR}

```

1: function TEMPORAL_DEDUCTIVE_REASONING( $D_{\text{relevant}}$ )
2:   Initialize  $A_{<t} \leftarrow \{\}$ ,  $E_{<t} \leftarrow \{\}$ 
3:   for each time point  $t$  in  $D_{\text{relevant}}$  do
4:      $D_t \leftarrow \text{FILTER}(D_{\text{relevant}}, \text{time} = t)$ 
5:      $A_t \leftarrow \text{SUMMARIZE\_ARTICLES}(D_t)$ 
6:      $E_t \leftarrow \text{EXTRACT\_EVENT\_TUPLES}(A_t)$ 
7:     Append  $A_t$  to  $A_{<t}$  and  $E_t$  to  $E_{<t}$ 
8:   end for
9:    $\text{analysis\_results} \leftarrow \text{LLM\_TEMPORAL\_ANALYSIS}(E_{<t}, A_{<t})$ 
10:   $\text{prompt}_{\text{TDR}} \leftarrow \text{CREATE\_PROMPT}(\text{analysis\_results})$ 
11:   $R_{\text{TDR}} \leftarrow \text{LLM\_PREDICT}(\text{prompt}_{\text{TDR}})$ 
12:  return  $R_{\text{TDR}}$ 
13: end function

```

Algorithm 2 Synchronic Associative Reasoning (SAR) Module

Require: Relevant news articles D_{relevant}
Ensure: Predicted relation set R_{SAR}

```

1: function SYNCHRONIC_ASSOCIATIVE_REASONING( $D_{\text{relevant}}$ )
2:   Group  $D_{\text{relevant}}$  into temporal intervals  $\{T_1, T_2, \dots\}$ 
3:    $P_{\text{all\_intervals}} \leftarrow \{\}$ 
4:   for each interval  $T_i$  do
5:      $D_{T_i} \leftarrow \text{GET\_ARTICLES\_IN\_INTERVAL}(D_{\text{relevant}}, T_i)$ 
6:      $D'_{T_i} \leftarrow \text{RANK\_AND\_SELECT\_ARTICLES}(D_{T_i})$ 
7:      $A'_{T_i} \leftarrow \text{SUMMARIZE\_TOP\_ARTICLES}(D'_{T_i})$ 
8:      $P_{T_i} \leftarrow \{\}$ 
9:      $\text{Source\_Pool} \leftarrow A'_{T_i}$ 
10:    for  $k \leftarrow 1$  to  $N_{\text{iterations}}$  do
11:       $\text{candidate\_inferences} \leftarrow \text{LLM\_GENERATE\_INFERENCES}(\text{Source\_Pool})$ 
12:       $\text{validated\_inferences} \leftarrow \text{LLM\_VALIDATE}(\text{candidate\_inferences}, \text{Source\_Pool})$ 
13:       $P_{T_i} \leftarrow P_{T_i} \cup \text{validated\_inferences}$ 
14:       $\text{Source\_Pool} \leftarrow A'_{T_i} \cup P_{T_i}$ 
15:    end for
16:    Append  $P_{T_i}$  to  $P_{\text{all\_intervals}}$ 
17:  end for
18:   $\text{prompt}_{\text{SAR}} \leftarrow \text{CREATE\_PROMPT}(P_{\text{all\_intervals}})$ 
19:   $R_{\text{SAR}} \leftarrow \text{LLM\_PREDICT}(\text{prompt}_{\text{SAR}})$ 
20:  return  $R_{\text{SAR}}$ 
21: end function

```

B Prompts

We provide the prompts for several key components.

Algorithm 3 Self-Correction (SC) Module

Require: Initial predicted relation set R_{initial}
Ensure: Corrected predicted relation set $R_{\text{corrected}}$

```

1: function SELF_CORRECTION( $R_{\text{initial}}$ )
2:    $\text{errors} \leftarrow \text{ANALYZE\_ERRORS}(R_{\text{initial}})$ 
3:    $R_{\text{corrected}} \leftarrow R_{\text{initial}}$ 
4:   for each  $\text{error}$  in  $\text{errors}$  do
5:     if  $\text{error.type} == \text{"FormatError"}$  then
6:        $R_{\text{corrected}} \leftarrow \text{CORRECT\_FORMAT}(R_{\text{corrected}}, \text{error})$ 
7:     else if  $\text{error.type} \in \{\text{"SemError"}, \text{"HallucError"}\}$  then
8:        $\text{prompt} \leftarrow \text{GEN\_CORRECTION\_PROMPT}(\text{error})$ 
9:        $R_{\text{corrected}} \leftarrow \text{LLM\_CORRECT}(\text{prompt}, R_{\text{corrected}})$ 
10:    end if
11:  end for
12:  return  $R_{\text{corrected}}$ 
13: end function

```

B.1 TDR Prompt

The prompt for conducting trend analysis after obtaining key news texts and historical events is as follows:

Trend Analysis Prompt

You are a seasoned expert in analyzing historical events and intelligence. The database contains news articles from January 1, 2023 to the current date and the events extracted from these articles. The events are in the form of (date, subject country, relation, object country), where the countries are represented by ISO 3166-1 alpha-3 codes and the relations are represented by the CAMEO codes defined in the 'Conflict and Mediation Event Observations' ontology. The relations are hierarchical: first-level relations are general parent relations represented by two-digit CAMEO codes, while second-level relations are more specific child relations represented by three-digit CAMEO codes. Child relations have the same first two digits as their parent relations. For example, '01' is a first-level relation, and '010' and '011' are some of its second-level relations. The relations in the database are represented in the second-level form.

Your task is to analyze useful intelligence based on some historical information that is already available. The information provided to you now includes a historical summary of events {events}. Distribution of all historical event relationships {relation_distributions}. And the specific relationship types corresponding to the top 20 most frequently occurring relation CAMEO codes {relations_description}.

Your task is to analyze the provided information to extract useful intelligence for future prediction. This involves several steps:

Evolutionary Trend Analysis: Based on the sequence of events, analyze the development and evolution patterns of the situation over time.

Distribution Analysis: Analyze the geographical and relational distribution of events, identifying key actors (countries) and frequent types of interactions.

Causal Relationship Analysis: Investigate the potential causal links between different events or a series of events. Determine what earlier events might have triggered subsequent events.

This information will be used to make predictions for the future, so please use it for this purpose and analyze as much as possible.

B.2 SAR Prompt

The prompt for generating new inferences based on available information is as follows:

Generate Inferences Prompt

You are a top AI scientist, logician, linguist, and intelligence analyst. Please think step by step.

The ‘news summaries’ you will be processing are sourced from a professional news database containing reports on international events from January 1, 2023, to the present. Each summary accurately describes a specific event, where the countries involved are represented by ISO 3166-1 alpha-3 codes, and the relations (interactions) between them are defined by codes from the CAMEO (Conflict and Mediation Event Observations) verb ontology. These summaries encapsulate key information, including the main countries involved, the nature of their interaction, and any important context or outcomes.

Based on the “summaries” of two news articles, use logical reasoning to derive a new “proposition” of intelligence. In other words, you need to reasonably deduce a new piece of information from the content of two news summaries.

Please ensure that the “proposition” is logically correct.

Please ensure that the “proposition” is not a repetition of the “summaries”.

Please ensure your reasoning is directly deduced from the “summaries” and does not introduce unsourced common knowledge or external information.

Remember, your “proposition” should help determine whether the “conclusion” is True, False, or Unknown.

Carefully study the following examples to understand how to generate a “proposition” from “summaries”. Each time, use only two “summaries” to generate one “proposition”. The proposition should help in judging the conclusion. The “explanation” in the examples is only for learning the reasoning process; do not provide explanations in the formal output.

Here are some examples:

{gen_proposition_examples}

This is the input for the proposition generation task:

“summaries”: {summaries}

Do not output the judgment of the conclusion. Strictly follow the example format and only present a single new proposition after “proposition:”. Do not include the “*” symbol, and do not repeat the “summaries” or “conclusion”. You need to learn the reasoning style from the examples, but do not output explanations or any other information in the formal output.

Do not repeat existing cases and prompts.

Please give your proposition: “proposition”:

Correction Prompt

You are a senior intelligence analyst responsible for the final review and correction of an AI model’s output. You are processing a large dataset of international events where each event has been classified using a CAMEO code.

Your task is to perform a systematic correction of the entire set of predicted results, guided by a pre-existing analysis of common errors.

You are now provided with two key inputs:

A detailed {error_summary}, which systematically outlines the typical error types and correction principles identified from the model’s earlier predictions.

The complete set of raw predictions that require correction, {total_prediction_results}.

Your mission is to meticulously review every single entry in {total_prediction_results}. Using the principles and insights from the {error_summary} as your guide, you must identify and correct all potential errors. You should generalize from the error summary, applying its logic to rectify similar mistakes, even if they aren’t explicitly identical to the examples in the summary.

Please output a new, fully revised version of the output. The format of your output should be identical to the input {total_prediction_results}, containing only the clean, corrected data.

C Description of Event Codes

The CAMEO ontology organizes event relations into a hierarchical structure, primarily divided into a binary classification of **Mediation** and **Conflict**. These are further categorized into four quadratic classes, each containing specific first-level relations defined by two-digit codes.

Under the **Mediation** category, *Verbal Cooperation* includes the following relations: “01: Make public statement”, “02: Appeal”, “03: Express intent to cooperate”, and “04: Consult”. The *Material Cooperation* class involves more tangible actions: “05: Engage in diplomatic cooperation”, “06: Engage in material cooperation”, “07: Provide aid”, and “08: Yield”.

The **Conflict** category is similarly subdivided. *Verbal Conflict* encompasses interactions such as: “09: Investigate”, “10: Demand”, “11: Disapprove”, “12: Reject”, “13: Threaten”, and “14: Protest”. Finally, *Material Conflict* describes severe or physical hostile actions: “15: Exhibit military posture”, “16: Reduce relations”, “17: Coerce”, “18: Assault”, “19: Fight”, and “20: Engage in unconventional mass violence”.

The ontology further refines these interactions into three-digit second-level codes to capture specific nuances. In this structure, the first two digits identify the parent relation, while the third digit specifies the precise nature of the interaction. For instance, the category “01: Make public statement” contains sub-events such as “010: Make statement, not specified”, “011: Decline comment”, “012: Make pessimistic comment”, etc.

D Details of Evaluation Metrics

Let $\mathcal{P}$ and $\mathcal{G}$ denote the sets of predicted and ground-truth relation codes. We evaluate performance using Precision (P), Recall (R), and F1-score:

B.3 SC Prompt

The prompt for using an LLM to correct the prediction results is as follows:

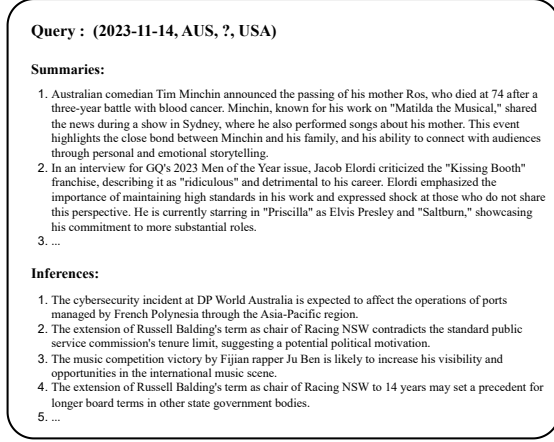


Figure 8: Summaries and inferences output.

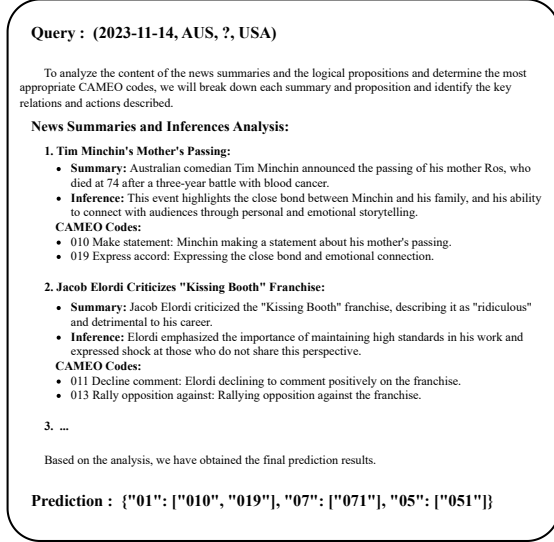


Figure 9: SAR analysis and prediction.

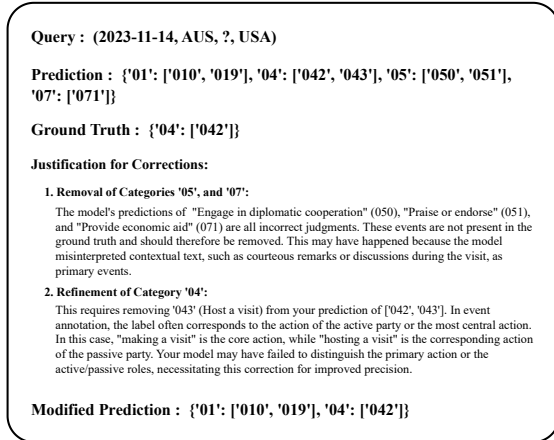


Figure 10: Self-correction process output.

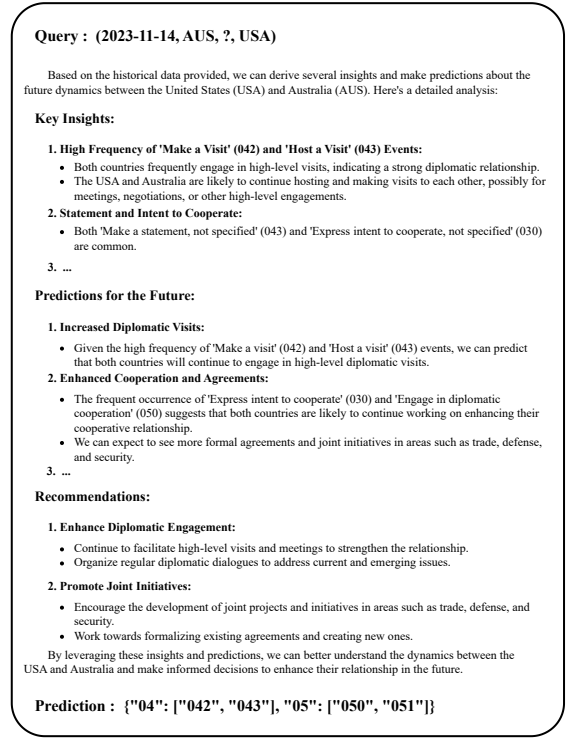


Figure 7: TDR analysis and prediction.

$$P = \frac{|\mathcal{P} \cap \mathcal{G}|}{|\mathcal{P}|}, \quad R = \frac{|\mathcal{P} \cap \mathcal{G}|}{|\mathcal{G}|}, \quad F_1 = 2 \cdot \frac{P \cdot R}{P + R}. \quad (1)$$

Furthermore, to measure the distributional discrepancy, we employ the Kullback-Leibler (KL) divergence:

$$D_{KL}(D_{\mathcal{G}} \parallel D_{\mathcal{P}}) = \sum_i D_{\mathcal{G}}(i) \log \left(\frac{D_{\mathcal{G}}(i)}{D_{\mathcal{P}}(i)} \right). \quad (2)$$

Here, $D_{\mathcal{G}}$ and $D_{\mathcal{P}}$ represent the frequency distributions of the ground-truth and predicted relations, respectively. A lower value of D_{KL} indicates a closer alignment between the predicted distribution and the ground truth.

E Case Study

We demonstrate our process with a case study predicting events between the United States and Australia for the query (2023-11-14, AUS, ?, USA). The outputs are presented sequentially: the TDR analysis (Figure 7), SAR reasoning (Figure 8), their combined result (Figure 9), and the final self-corrected output (Figure 10). This process yields the final prediction: “{‘01’: [‘010’, ‘019’], ‘04’: [‘042’]}”.